

LEBANESE
INTERNATIONAL
UNIVERSITY



الجامعة اللبنانية الدولية
Lebanese International University

Lebanese International University
School of Arts and Sciences
Department of Mathematics and Physiques

Summer 2023-2024

MATH310 – Probability and Statistics

Instructor:
E-Mail: @liu.edu.lb
Campus:
Lectures time:

Course Objectives

This course aims to:

- Provide students with the basic probabilistic and statistical concepts with related computational and analytic skills.
- Provide student an adequate ability for understanding and interpreting many non-deterministic situations.
- Develop an understanding the wide range of applications of probabilistic and statistical to real life situations.

Student's Learning Outcomes

Upon successful completion of this course, the student should be able to:

1. Develop skills and knowledge of factorial and the basic principal of counting.
2. Use Conditional probability, Rule of total probability and Bayes' theorem.
3. Identify the concept of a random variable and the probability distributions.
4. Explain the expected value, variance and covariance of a random variable.
5. Evaluate all relevant statistics from a sample, practice sampling procedures and interpret descriptive statistical displays.
6. Construct and interpret confidence intervals and hypothesis testing for estimating population parameters.

Textbook and Recommended Material

Textbook

Probability and Statistics for Engineers and Scientists, Walpole and Myers, 9th Ed., Prentice Hall

Policies and procedures: Students caught cheating on an exam receives a grade of **F** on the exam in their first cheating attempt and receives a warning. Students caught cheating for the second time will receive a grade of “**F**” on the course.

Grading scheme:

Evaluation	Date	Weight	Chapters	CLO
Exam 1	TBA	27%	1, 2, 3.1	1 and 2
Exam 2	TBA	27%	3.4, 4, 5	3, 4, 5
Attendance		6%		
Final	TBA	40%	6, 8, 9	5 and 6

Attendance: Attendance will be recorded at the beginning of each session. You are allowed to be late up to five minutes only, regardless of your excuses. If you are late more than five minutes you will be considered absent. So try hard to be always on time.

Missing exam: There will be makeup exam if you have a valid excuse according to LIU rules and regulation. After approval of the exam petition, you will be given a makeup exam at an arranged time decided by the administration and the instructor. The Make-up exam will be marked over 80 percent of the missed exam.

Responsibilities: Students must read the chapters and work out all assigned problems given in the handout. These problems should be handed in to the instructor after each chapter is done to avoid accumulation before each midterm. Understanding and working out problems is very crucial for passing the course. In addition, students are strongly encouraged to meet with the instructor during office hours for further questions and explanation of any struggled issues.

Solved problems set: In the table below there is a list of problems toward every chapter solved by the instructor in class. In addition, you will have a set of problems and previous exams with solution sent to you by e-mail and provided at the copy centers. These sets will be of big help for studying and preparation for the exams.

Course Schedule

weeks	Chapter	Sections and required exercise to solve	HW
Week 1 (2 lectures)	Chapter 2: Probability	2.3 Counting Sample Points Examples: 2.3,2.13,2.15,2.17-2.19, 2.21-2.23 Exercises: 2.32,2.37,2.38 p.51-52	Exercises: 2.21,2.26,2.30, 2.33,2.34,2.36, 2.43 p.51-52
Week 1 (2 lectures)		2.4-2.6 Probability Examples: 2.25,2.27-,2.30,2.34,2.36-2.40 Exercises: 2.53,2.75 pages:59 and 69	Exercises: 2.53,2.58,2.61, 2.72,2.83 ,2.89 p.60,70
Week 2 (1 lecture)		2.7 Bayes' Rule Examples: 2.41-2.43 p.74-76	Exercises: 2.96,2.98,2.99 p.76-77
Week 2 (2 lectures)	Chapter 3: Random Variables and Probability Distributions	3.1-3.3 One Random Variable Examples 3.8,3.10,3.11,3.12 p.85-90 Exercises 3.12, 3.21 p.92-93	Exercises: 3.7, 3.9,3.14,,3.26,3. 27,3.29,3.30 p.92-93
Week 2-3 (2 lectures)		3.4 Two Random Variables Examples: 3.14-3.21 Exercises: 3.44, 3.49	Exercises: 3.38, 3.41,3.42,3.43,3. 44,3.46,3.49,3.5 0 p.104-106
Week 3 (1.5 lectures)	Chapter 4: Mathematical Expectation	4.1-4.3 Mean Variance and Covariance of a Random Variable Examples: 4.1, 4.3-4.14 Exercises: 4.17, 4.20, 4.58 Example 4.22, 4.23 p.133 Exercises: 4.59, 4.64 p.138	Exercises: 4.23, 4.41,4.52 4.60, 4.62, 4.70
Week3 (1 lecture)	Chapter 5: Some Discrete Probability Distributions	5.2 Binomial Distributions Example 5.1,5.25.3-5.6 p.145-148	Exercises: 5.6,5.10,5.15 p.150-151
Week 4 (1 lectures)		5.5 Poisson Distribution (without approximation Binomial by Poisson) Example:5.17,5.18,5.19,5.20 p.162-164	Exercises: 5.56,5.57,5.61, 5.62, p.165
Week 4 (1.5 lectures)	Chapter 6: Some Continuous Probability Distributions	6.2-6.4 Normal Distribution Examples: 6.2,6.3,6.4, 6.6,6.7,6.8,6.10,6.13 p.178-185	Exercises: 6.8, 6.9, 6.11,6.14, 6.16 p.186-187
Week 4 (1 lecture)	Chapter 8: Sampling Distributions	8.4 Sampling Distribution of Means and the Central Limit Theorem Examples: 8.4, Case study: 8.2 and 8.6 Exercises: 8.25,8.26 p.242	Exercises: 8.23, 8.30,8.32
Week 5 (1 lectures)		8.5 Chi-squared distribution Extra exercises	Exercises: 8.39, 8.46,8.47, 8.48 p.259
		8.6 t-Distribution Examples: 8.8,8.9,8.10 p.249	

Week 5 (1 lecture)	Chapter 9: One Sample Estimation Problems	9.4 Single Sample: Estimating the Mean Examples: 9.2,9.3,9.5,9.6 p.271-276	Exercises: 9.2,9.3, 9.4,9.6,9.7, 9.11, 9.12
Week 5 (1/2 lecture)	Chapter 9: One Sample Estimation Problems	9.10 Single Sample: Estimating a Proportion Examples: 9.14,9.15,9.16 p.297-299	Exercises: 9.51, 9.53, 9.56, 9.59, 9.60 p.302
Week 5 (1/2 lecture)	Chapter 9: One Sample Estimation Problems	9.12 Single Sample: Estimating the Variance Examples: 9.18 p.304	Exercises: 9.71,9.72 p.307